1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In adenocarcinomas with BRAF V600E and PTEN loss, a high stromal-to-tumor ratio (TSR) will be associated with reduced therapy_response to MAPK/BRAF-targeted therapies. This is because PTEN loss sustains PI3K/AKT signaling, promoting tumor cell survival despite MAPK inhibition, and a stromal-rich microenvironment (high TSR) creates physical and biochemical barriers that limit drug penetration and foster resistance. Consequently, high-TSR tumors with PTEN loss and BRAF V600E will demonstrate poorer therapeutic responses compared with low-TSR counterparts. This suggests that, in such tumors, combining MAPK/BRAF-targeted therapy with PI3K/AKT–pathway–modulating strategies or stromal-normalizing approaches should improve therapy_response. The hypothesis is testable by correlating TSR with therapy_response in PTEN-loss/BRAF V600E tumors across model systems and, where possible, clinical datasets, and by evaluating whether co-targeting the PI3K/AKT axis or altering the tumor microenvironment enhances response in high-TSR contexts.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PTEN, BRAF
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTEN: PTEN perturbation and BRAF activating mutations in adenocarcinoma contexts, raising the question: why do BRAF-targeted interventions sometimes fail when PTEN is altered? 2) Tier 1 path(s) used: the single path linking adenocarcinoma → PTEN (PTEN loss/perturbation) and the presence/therapeutic targeting of BRAF V600E (BRAF activating mutation).

BRAF: BRAF activating mutations in adenocarcinoma contexts, raising the question: why do BRAF-targeted interventions sometimes fail when PTEN is altered? 2) Tier 1 path(s) used: the single path linking adenocarcinoma → PTEN (PTEN loss/perturbation) and the presence/therapeutic targeting of BRAF V600E (BRAF activating mutation).

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions